

Nanotribology of Ionic liquids as Lubricant Additives for Alumina Surfaces

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Supplemental Information

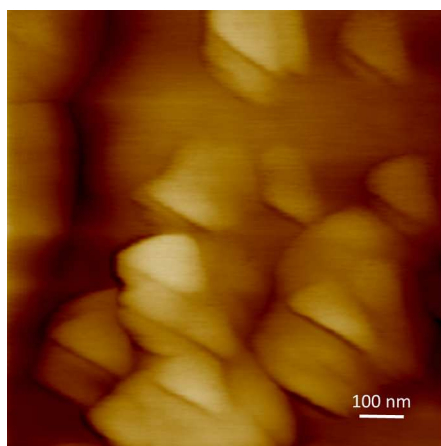


Figure S1. AFM image of Alumina Surface

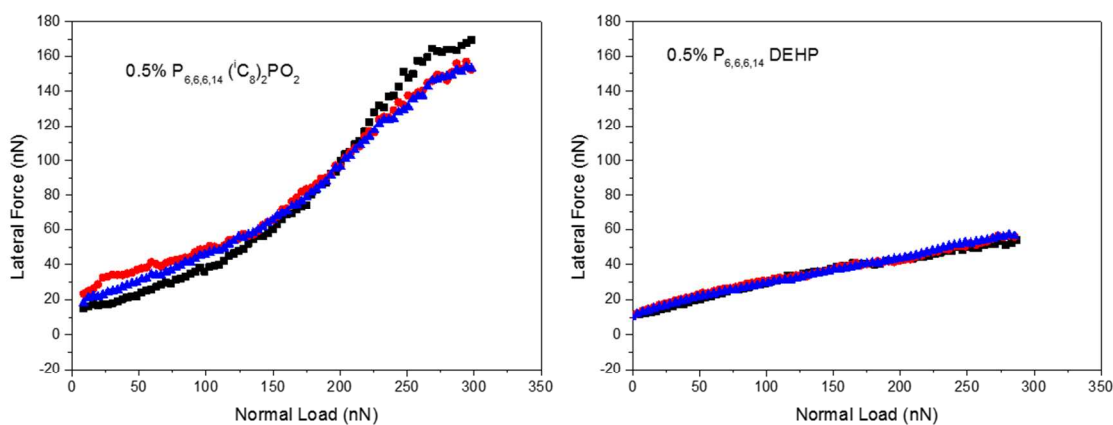


Figure S2. Repeated friction curves for 0.5 mol% P_{6,6,6,14}(C₈)₂PO₂/hexadecane and 0.5 mol% P_{6,6,6,14} DEHP/hexadecane mixtures

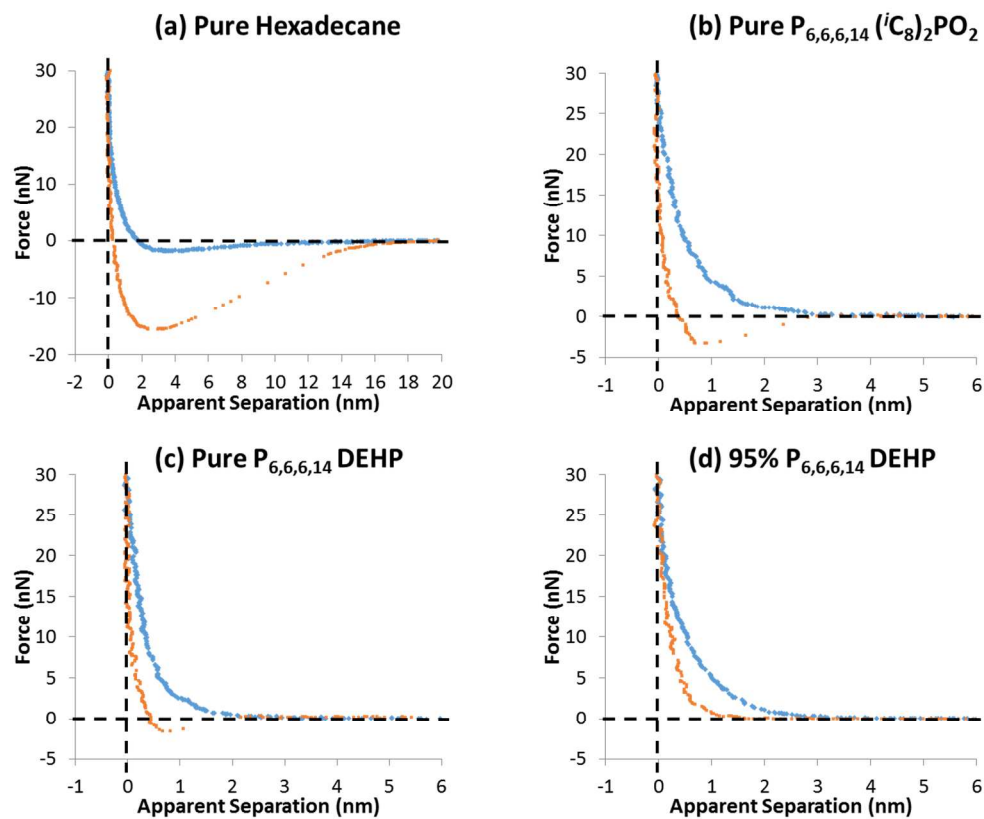


Figure S3. Typical normal force versus apparent separation profile for an AFM tip approaching and retracting from an alumina surface in pure hexadecane (a), pure $P_{6,6,6,14} (iC_8)_2PO_2$ (b) and $P_{6,6,6,14}$ DEHP (c) and 95 mol% $P_{6,6,6,14}$ DEHP/hexadecane mixture (d).

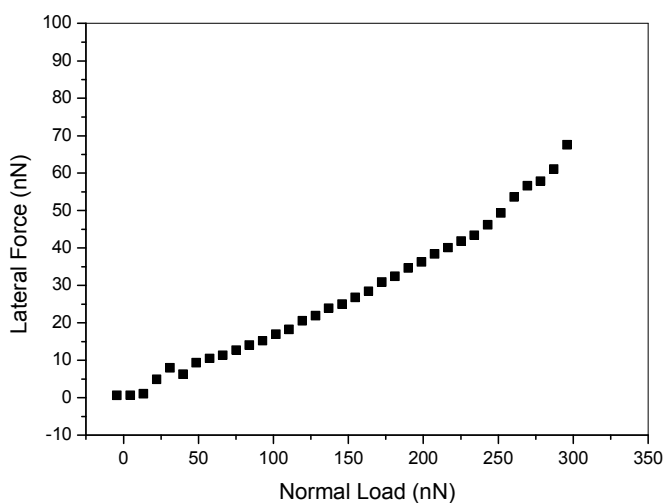


Figure S4. Lateral force vs normal load for 95% P_{6,6,6,14} DEHP in Hexadecane.

Table S1. Zero shear viscosity of P_{6,6,6,14} (C₈)₂PO₂/hexadecane mixtures at 20 °C measured by an AR-G2 rheometer (TA instruments).

	0.1 mol%	0.5 mol%	1.0 mol%	2.0 mol%	5.0 mol%
	IL				
Viscosity(mPa s)	3.9	3.9	4.0	4.2	4.9

Table S2. Zero shear viscosity of P_{6,6,6,14} DEHP /hexadecane mixtures at 20 °C measured by an AR-G2 rheometer (TA instruments).

	0.01	0.03	0.1	0.5	1.0	2.0	5.0
	mol%	mol%	mol%	mol%	mol%	mol%	mol% IL
Viscosity(mPa s)	3.8	3.8	3.9	4.0	4.0	4.3	4.9